Damon H. T. Cheung

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EDUCATION

University of Michigan, Ann Arbor GPA: 3.8/4.0
Ph.D. in Physics 2022 – Present

The Chinese University of Hong Kong
B.Sc. in Physics — *Enrichment Stream in Theoretical Physics* 2018 – 2022

AWARDS & LEADERSHIP

- LIGO-Virgo-KAGRA Continuous Waves Working Group, Directed Searches Subgroup Chair 2026 – Present
- Frank Sevcik Award 2026
- Michigan Institute for Computational Discovery & Engineering Fellowship 2025

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Michigan, Ann Arbor Sep 2022 – Present

- Developing a semicoherent matched-filtering search pipeline for continuous gravitational-wave signals from binary systems using JAX.
- Engineered an Attention U-Net deep learning model to extract low-SNR signals from noisy time-series data, matching state-of-the-art detection sensitivity with less than half the training time and data.
- Developed a modular Python package to automate multi-target continuous gravitational-wave searches on large-scale datasets.

AI/ML Research Intern, Polymathic Technologies Inc., New York May 2026 – Aug 2026

- Conducted research and development on multimodal foundation models using JAX, expanding multimodal capabilities and improving downstream task performance.
- Investigated the performance of time-series tokenization on real-world data and integrated it with a foundation model for time-series prediction.

Research Assistant, The Chinese University of Hong Kong Dec 2021 – Aug 2022

- Derived the correct form of a Bayesian hypothesis test for rare events by reweighting with the population prior, and showed that nonparametric density estimation reduces prior-induced bias relative to parametric models.

Guest Researcher, Center for Computational Astrophysics, Flatiron Institute May 2021 – Dec 2021

- Implemented hierarchical Bayesian inference with accelerated likelihood emulators built from Gaussian processes combined with Principal Component Analysis (PCA) and masked autoregressive normalizing flows.

SELECTED PUBLICATIONS

*Author of **50+** papers with **4,000+** citations and an h-index of 26. Full list available on INSPIRE.*

- **The G347.3-0.5 outlier from O3: a follow-up case study for continuous gravitational-wave candidates**, arXiv:2608.07662 (2026)
  Lorenzo Mirasola, Francesco Amicucci, Marta Carrió Thomás, **Damon H. T. Cheung**, *et al.*
- **Searches for Continuous Gravitational Waves from Supernova Remnants in the first part of the LIGO-Virgo-KAGRA Fourth Observing run**, ApJ 1004, 204 (2026)
  A. G. Abac, *et al.* (**Damon H. T. Cheung** as a main contributor)
- **Search for continuous gravitational waves from neutron stars in five globular clusters in the first part of the fourth LIGO–Virgo–KAGRA observing run**, Phys. Rev. D 114, 062001 (2026)
  Damon H. T. Cheung, Keith Riles, *et al.*
- **Attention U-Net for all-sky continuous gravitational wave searches**, Phys. Rev. D 112, 103039 (2025)
  Damon H. T. Cheung
- **Mitigating model uncertainty on the strong lensing Bayes factor using nonparametric methods**, Phys. Rev. D 111, 063012 (2025)
  Damon H. T. Cheung, Stefano Rinaldi, *et al.*
- **Testing the robustness of simulation-based population inference**, Phys. Rev. D 106, 083014 (2022)
  Damon H. T. Cheung, Kaze W. K. Wong, *et al.*